

William Kojumian

Autonomous Systems | Robot Task Planning | Machine Perception

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🌐 github.com/twillur — [🌐 Research blog](#) — 📍 Dubai, UAE

Research Interests

Autonomous mobile robotics, robot task planning, and machine perception. My current work sits at the point where human instructions must become machine-executable robot behaviour — how natural language is grounded into verifiable plans, and how the reliability of that translation can be measured rather than asserted. I am particularly interested in the evaluation problem: establishing where a learned or language-driven component stops being trustworthy, and designing systems that recognise and declare their own limits.

Education

MSc Robotics Engineering **Sep 2025 – Dec 2026**
Middlesex University Dubai (Middlesex University London), UAE *Expected: Distinction*
Completed taught modules: 120 of 180 credits, distinction-level average
Dissertation submitted October 2026; degree completion expected December 2026
Robotic Systems Integration · Sensing & Motion Control · Robot Manipulation · Mobile Robotics · Machine Learning for Engineers · Engineering Sustainability

BSc Engineering Science (Mechatronics) — four-year programme **Sep 2021 – Jun 2025**
American University of Armenia, Yerevan
Circuits · Embedded Systems · Electromechanics · Control Theory · CAD · Programming

American High School Diploma **2020 – 2021**
Al Ma'arifa International School, UAE

Research

MSc Dissertation — Natural Language Task Planning for Autonomous Ground Robot Navigation via Large Language Models **2026**
Supervisor: Dr Judhi Prasetyo, Middlesex University Dubai Use case: warehouse patrol & inspection

- **Research question:** how reliably can a prompt-engineered large language model translate natural-language commands into structured Nav2 waypoint plans, across five levels of linguistic and spatial complexity?
- **Method:** designed a schema-constrained plan representation so that model output is machine-verifiable rather than free text; built a semantic environment map to ground references to named locations; assembled a stratified evaluation dataset of 100 commands across five complexity levels, including deliberately unsatisfiable commands that test whether the planner correctly *refuses* instead of producing a confident but wrong plan.
- **Implementation:** provider-agnostic planner and evaluation harness, integrated end to end with ROS2 Humble, Nav2 and Gazebo.
- **Results:** 96.7%, 95.0% and 95.0% at complexity levels 1–3, falling to 55.0% at level 4 (conditional) and recovering to 85.0% at level 5 (ambiguous). Saved level-4 errors traced to a missing alternative-destination field in the plan schema; adding it raised schema validity from 55/60 to 60/60 and fallback-branch encoding from 0/9 to 7/9 (4/6 excluding the prompt-example command). Failure modes, including a fallback that failed in Gazebo execution, were documented rather than omitted.
- **Contribution:** the prompt architecture together with the graded evaluation methodology — the

framework is what makes claims about the system falsifiable, rather than the integration itself. Code: [Twillur/PDE4445-LLM-Nav2-Planner](#)

Public research documentation

2026

- Maintained a dated, week-by-week public record of the project including negative results, failed experiments and design reversals: [twillur.github.io/PDE4445-Robotics-Dissertation](#)

Engineering Projects

Sandsifter — Autonomous Beach Cleaning Robot

2025–2026

- Full ROS 2 (Humble) and Gazebo Classic 11 simulation of an autonomous beach-cleaning robot, integrating a `ros2.control` manipulator arm for debris collection.
- Implemented a custom boustrophedon coverage path planner for complete-area traversal, with sensor-driven obstacle avoidance.

Visual Quality Inspection — Tissue Box Classifier

2025

- Fine-tuned **MobileNetV2** via transfer learning on a self-collected image dataset to classify packaging as PASS or FAIL, reaching ~95% validation accuracy; built the full pipeline from data collection and augmentation through training and evaluation. [Twillur/PDE4444-Visual-Quality-Inspection](#)

Epson RC+ Industrial Robot Automation

2025

- Programmed an Epson industrial arm for autonomous pick-and-place with sorting logic, integrating sensor feedback for object detection and trajectory refinement.

BSc Capstone — Drone System for Intralogistical Payload Transport

2024–2025

- Designed and prototyped a UAV system for autonomous payload transport, covering flight dynamics, microcontroller integration and navigation logic. Two-semester degree project, published by AUA: [cse.aua.am](#)

Professional Experience

AI Engineering Intern

Mar 2026 – Jul 2026

Enhance Intelligence & Technologies | Dubai, UAE

- Built **IntelliChat**, a production retrieval-augmented assistant translating natural language into Oracle SQL over live ERP data across nine business domains, with role-based access and multi-tenant deployment (Blazor Server, .NET/C#).
- Engineered a semantic query cache using **Oracle 23ai** vector embeddings with cosine similarity search, and vector-based domain detection that cut prompt size 50–60% per request; integrated a local embedding model (nomic-embed-text-v1.5) to remove embedding API cost.

Technical Lead & Club Organizer

2021 – 2025

Drone Club, American University of Armenia | Yerevan, Armenia

- Led UAV workshops across four years covering flight physics, assembly, simulation workflows and flight training; directed student teams through build-and-test missions.

Additional Activities

- **Robotics demonstrator, Doer's Summit 2025** (three-day event, Dubai Silicon Oasis; Intel-sponsored) — operated Unitree humanoid and quadruped platforms in live public demonstrations.

Technical Skills

- **Robotics:** ROS 2 (Humble), Nav2, Gazebo Classic, `ros2_control`, behaviour trees, motion control, sensor integration, embedded systems, UAV systems, Epson RC+, Fusion 360
- **AI & Machine Learning:** PyTorch, transfer learning, LLM integration and prompt architecture (GPT-4o, OpenAI API), retrieval-augmented generation, vector databases, semantic search, LangChain, empirical model evaluation
- **Programming & Tools:** Python, C++, C# (.NET / Blazor Server), MATLAB, Git, Oracle 23ai, SQL, $\LaTeX$, Linux/Ubuntu, Docker (basic)
- **Languages:** English (fluent — full medium of instruction, BSc and MSc), Arabic (professional), Armenian (professional)